

Modeling Canada Housing Prices

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Introduction

In this pdf, I will using R to attempt to model the response variable housing price based on multiple explanatory variables:

- Bedrooms: Number of bedrooms in the property (main or otherwise)
- Bathrooms: Number of bathrooms in the property (main or otherwise)
- Square Feet: Rough square footage of indoor property
- Property Type: Category assigned to each listing including...
 - Apt/Condo
 - Duplex
 - Fourplex
 - House
 - Land/Lot
 - Mfd/Mobile Home
 - Multifamily
 - Recreational
 - Townhouse
 - Triplex

Forming the Initial Model

Let's use some libraries first:

```
library(tidyverse)
library(ggsci)
```

Then we can import the raw scraped housing data that was collected beforehand:

```
# Read in csv file as a tibble,
houses.df <- read_csv(file = "housing_df.csv", show_col_types = FALSE)
# remove any data with invalid values,
houses.df <- houses.df %>%
  drop_na()
# get rid of duplicates (some listings appeared multiple times),
houses.df <- houses.df %>%
  distinct()
# make Property Type a factor, and order it
houses.df$`Property Type` <- factor(houses.df$`Property Type`)
houses.df <- houses.df %>%
  mutate(`Property Type` = fct_reorder(`Property Type`, Price,
                                         .fun="length", .desc = TRUE))
```

Let's see an initial naive model:

```
houses.glm1 <- glm(formula = Price~Bedrooms+Bathrooms+`Square Feet`+`Property Type`,
                    data=houses.df)
summary(houses.glm1)
```

```
##
## Call:
## glm(formula = Price ~ Bedrooms + Bathrooms + `Square Feet` +
##      `Property Type`, data = houses.df)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -7387707  -276806   -61080   164687  25426804
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.530e+05  1.890e+04  -8.097 6.23e-16 ***
## Bedrooms       -5.873e+04  5.327e+03 -11.025 < 2e-16 ***
## Bathrooms       7.103e+04  7.283e+03  9.752 < 2e-16 ***
## `Square Feet`   5.935e+02  9.795e+00  60.599 < 2e-16 ***
## `Property Type`Apt/Condo    1.956e+05  1.616e+04  12.104 < 2e-16 ***
## `Property Type`Townhouse   -2.004e+04  1.744e+04  -1.149 0.250443
## `Property Type`Duplex       1.210e+05  3.017e+04   4.009 6.14e-05 ***
## `Property Type`Mfd/Mobile Home -2.262e+05  6.139e+04  -3.685 0.000229 ***
## `Property Type`Multifamily  -9.798e+04  7.403e+04  -1.323 0.185698
## `Property Type`Fourplex    -3.223e+05  1.828e+05  -1.763 0.077949 .
## `Property Type`Land/Lot    -1.412e+05  1.914e+05  -0.738 0.460778
## `Property Type`Triplex     -1.040e+05  2.019e+05  -0.515 0.606561
## `Property Type`Recreational -3.026e+05  3.492e+05  -0.867 0.386211
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 365468683316)
##
##      Null deviance: 7.9735e+15  on 11327  degrees of freedom
## Residual deviance: 4.1353e+15  on 11315  degrees of freedom
## AIC: 333764
##
## Number of Fisher Scoring iterations: 2
```

Remarks on initial model:

- the Bedrooms variable is oddly assigned a negative (and significant) correlation with Price, which is counter-intuitive
- multiple Property Type categories are deemed to have insignificant difference from “House”

Refining the Dataframe

Before creating another model, I will be going through each variable (response or explanatory) to refine our dataframe. This includes graphing their distributions and inspecting specific qualities to remove outlying/incorrect data and develop an deeper understanding about our dataset.

Checking Price

Let's check the distribution of house listing prices, to see if we can remove any outliers.

```
#' Creates quantiles of middle 99% range of data and distribution plot
#'
```

```
#' @param df A Dataframe
#'
```

```
#' @param column A string representing a column name
#'
```

```
#' @param datatype A string denoting plot type
#'
```

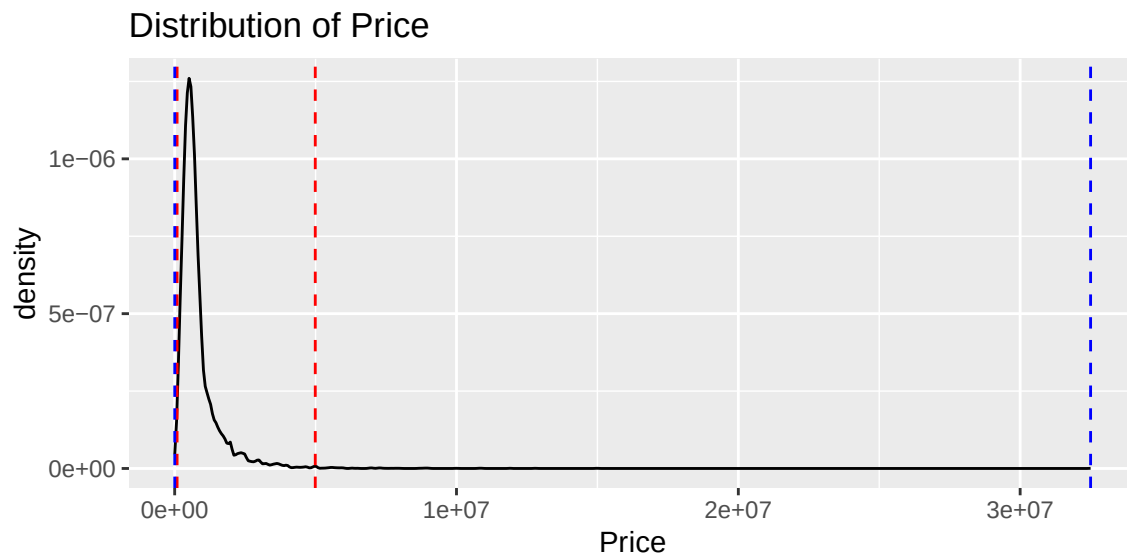
```
#' @return list of quantile values
#'
```

```
#' @return distribution plot
```

```
distribution_graph <- function(df, column, datatype="bar") {
  q <- quantile(df[[column]], probs=c(0,0.005,0.995,1))
  y <- if(datatype=="density"){geom_density()}else{geom_bar()}
  p <- ggplot(data=df, aes(x=.data[[column]])) + y +
    ggtitle(paste("Distribution of", column)) +
    geom_vline(xintercept=q[c(2,3)], linetype="dashed", color="red") +
    geom_vline(xintercept=q[c(1,4)], linetype="dashed", color="blue")
  return(list(quantiles=q,plot=p))
}
```

```
distribution_graph(houses.df, "Price","density")
```

```
## $quantiles
##      0%      0.5%     99.5%    100%
##    3100    85000  4988730 32500000
##
## $plot
```

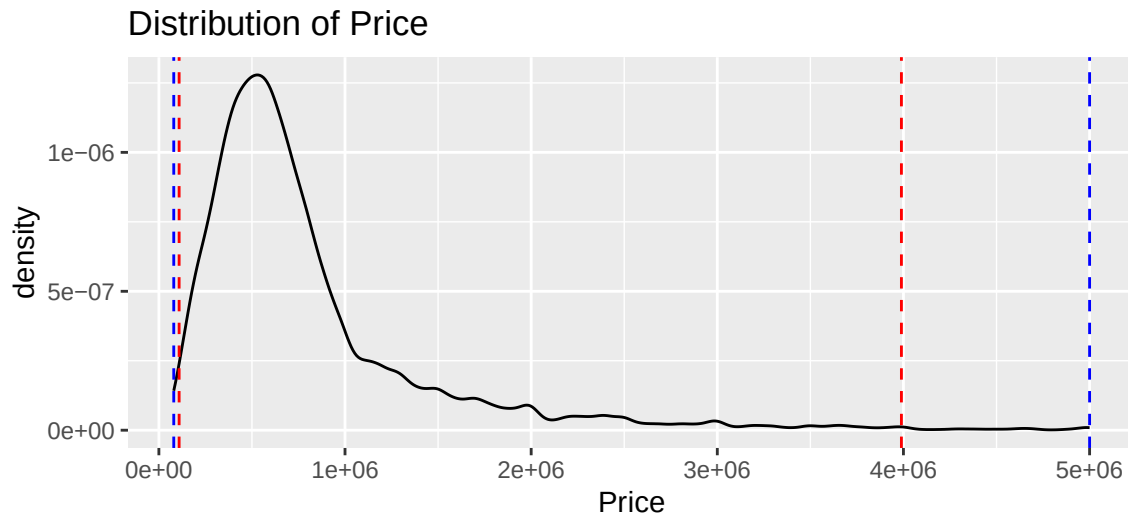


This distribution is a little too skewed by some extremely expensive house listings. Since 99% of the price values lie within the prices of \$80,000 to \$5,000,000, let's remove the values which go beyond this range to make the graph more normally distributed.

```
# filter out outlying prices and redo plotting:
houses.df <- houses.df %>%
  filter(Price <= 5000000) %>%
  filter(Price >= 80000)
price.plot <- distribution_graph(houses.df, "Price","density")
price.plot$quantiles
```

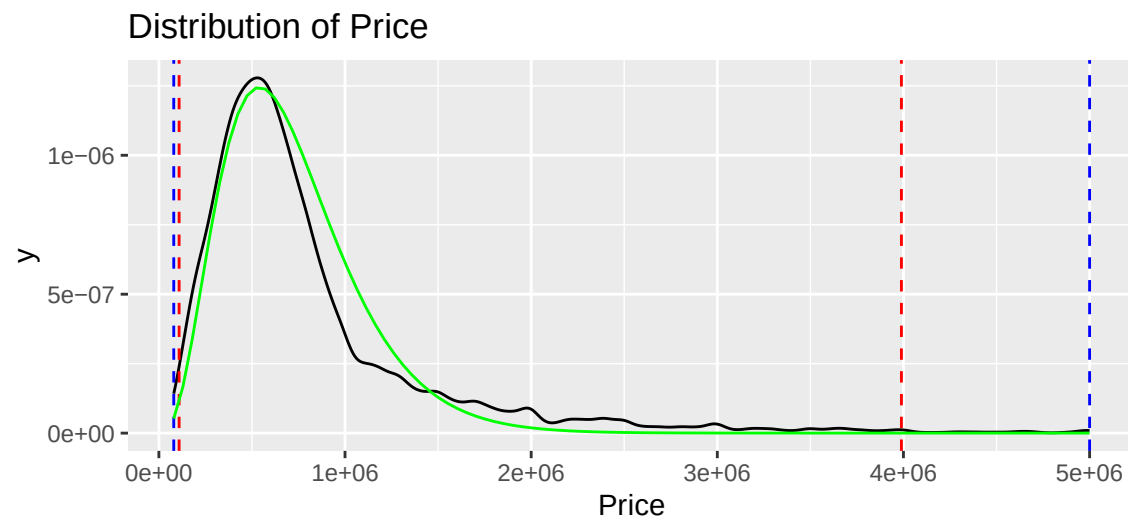
```
##          0%          0.5%          99.5%          100%
## 80000.0 108729.1 3988984.5 5000000.0
```

```
price.plot$plot
```



This distribution is now a lot more normal for our model to handle. However the right-skewedness suggests that a gamma distribution would be more fit to model Price. Using a gamma shape of 4 seems to generally fit our model, and ensures that Price remains positive:

```
price.plot$plot +
  stat_function(fun=dgamma, args=list(shape=4,scale=180000),
    color="green", show.legend = TRUE)
```

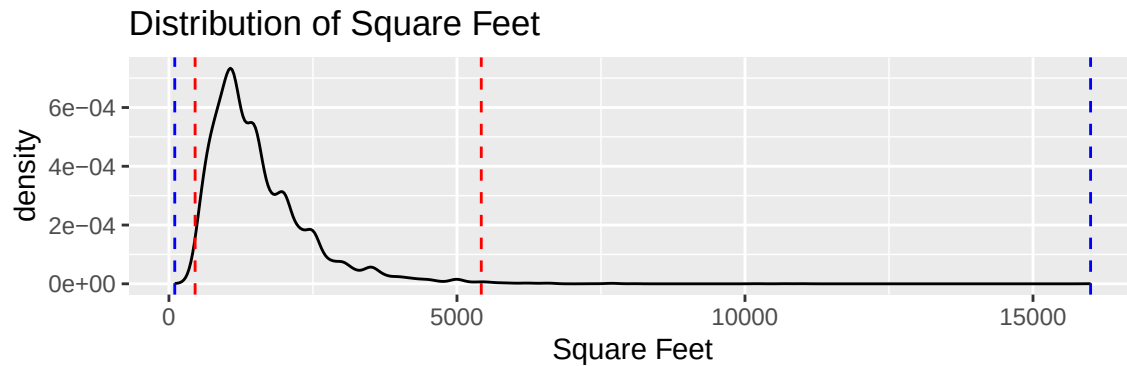


Checking Square Feet

Let's continue by checking the distribution of house square footage.

```
distribution_graph(houses.df, "Square Feet", "density")
```

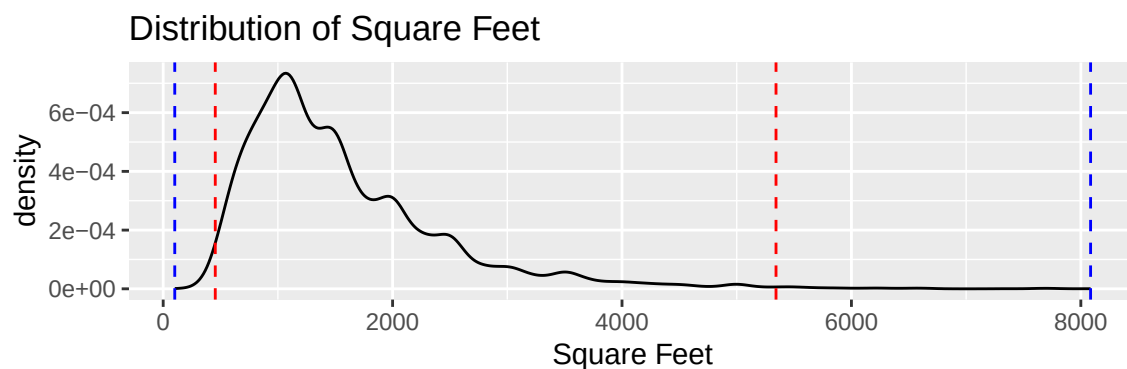
```
## $quantiles
##      0%  0.5% 99.5% 100%
##    101   454  5422 16000
##
## $plot
```



Since this distribution already seems relatively normal, I will only remove listings above 10,000 square feet.

```
# filter out outlying Square Feet and redo plotting:
houses.df <- houses.df %>%
  filter(`Square Feet` <= 10000)
distribution_graph(houses.df, "Square Feet", "density")
```

```
## $quantiles
##      0%   0.5% 99.5% 100%
## 101.00 454.00 5342.46 8085.00
##
## $plot
```



Checking Bathrooms

Now moving onto bathrooms, which seem normal already:

```
distribution_graph(houses.df, "Bathrooms")
```

```
## $quantiles
##    0%  0.5% 99.5% 100%
##     0     1     7    12
##
## $plot
```

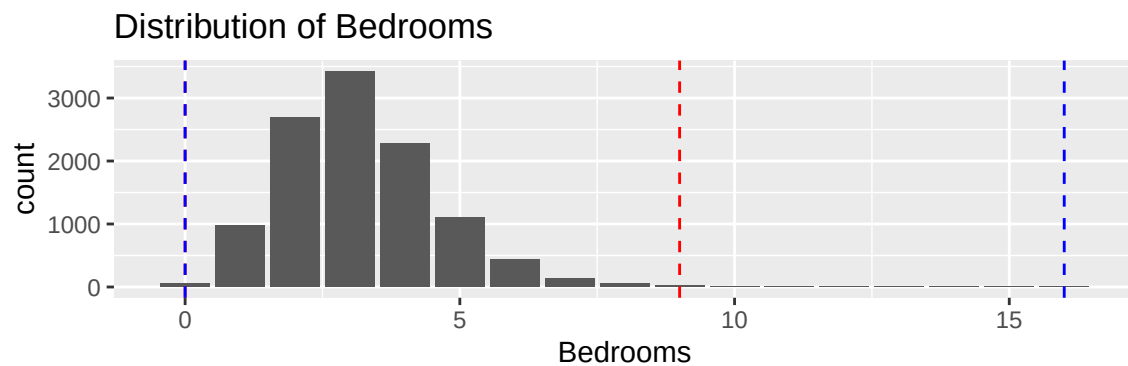


Checking Bedrooms

Finally bedrooms, which also seems normal already:

```
distribution_graph(houses.df, "Bedrooms")
```

```
## $quantiles
##    0%  0.5% 99.5% 100%
##     0     0     9    16
##
## $plot
```

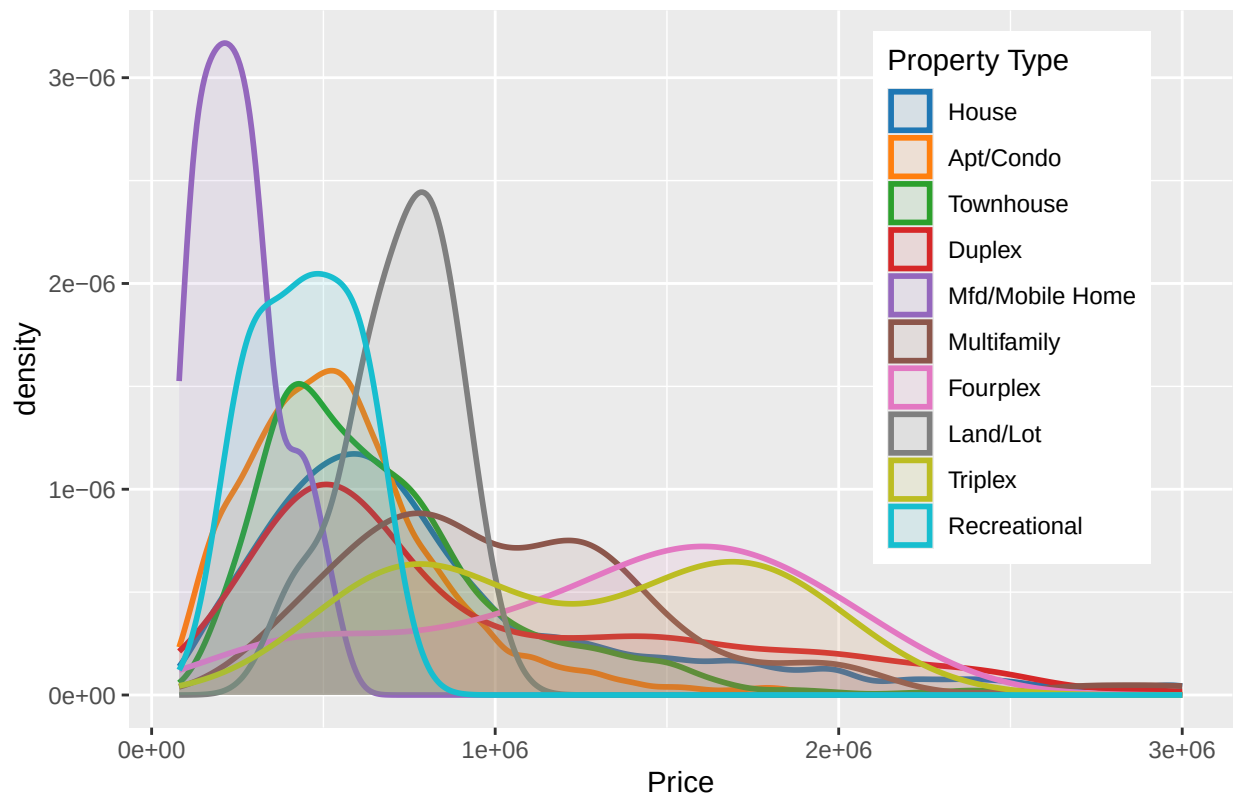


Checking Property Type

Now let's look at the distributions of the categorical variable property types:

```
houses.df %>%  
  group_by(`Property Type`) %>%  
  summarise(count=n())  
  
## # A tibble: 10 x 2  
##   `Property Type` count  
##   <fct>          <int>  
## 1 House          5951  
## 2 Apt/Condo      2970  
## 3 Townhouse      1688  
## 4 Duplex         436  
## 5 Mfd/Mobile Home  83  
## 6 Multifamily     68  
## 7 Fourplex       11  
## 8 Land/Lot        9  
## 9 Triplex         9  
## 10 Recreational   3  
  
ggplot(data=houses.df[houses.df$Price<=3000000,],  
  aes(x=Price, fill=`Property Type`, color=`Property Type`)) +  
  geom_density(linewidth=1,alpha=0.1) + ggtitle("Price Boxplot for Each Property Type") +  
  theme(legend.position="inside", legend.position.inside = c(0.8, 0.6)) +  
  scale_color_d3() + scale_fill_d3()
```

Price Boxplot for Each Property Type



*Note: x-axis has been shrunk to make distributions clearer

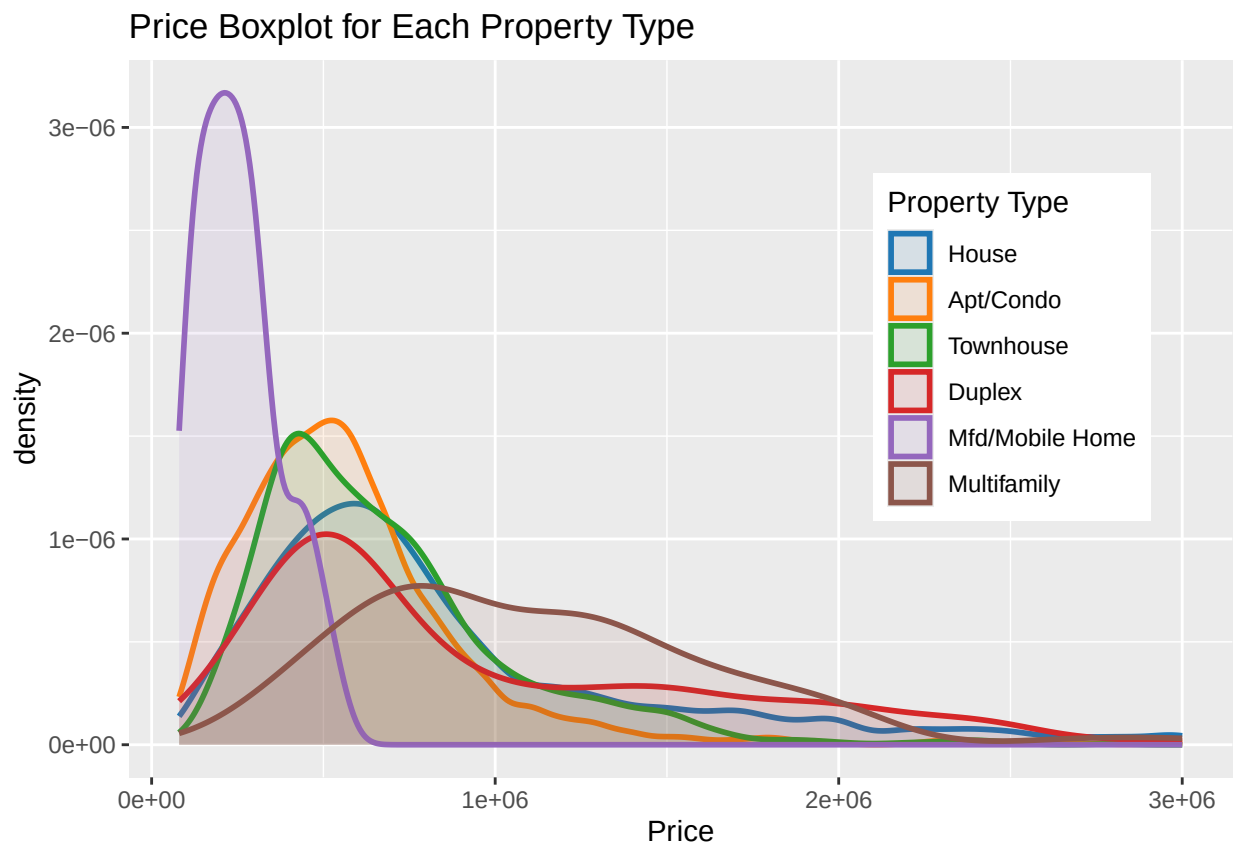
Due to small sizes of some categories, I will be merging some categories into one based on similar distribution/property type. First I will merge “Multifamily”, “Triplex”, and “Fourplex”, since each has a wider further-right distribution, and are all properties with multiple living units. Since “Recreational” and “Land/Lot” both have so few occurrences combined with unique distribution and property type, I will be removing these listings from our dataset:

```
# multi-family merging
houses.df[houses.df$`Property Type` %in%
  c("Triplex", "Fourplex"),]$`Property Type` <-
  "Multifamily"

# removing categories
houses.df <- houses.df %>%
  filter(!(`Property Type` %in% c("Land/Lot", "Recreational")))
```

Now let's recheck the distributions:

```
ggplot(data=houses.df[houses.df$Price<=3000000,],
  aes(x=Price, fill=`Property Type`, color=`Property Type`)) +
  geom_density(linewidth=1,alpha=0.1) + ggtitle("Price Boxplot for Each Property Type") +
  theme(legend.position="inside", legend.position.inside = c(0.8, 0.6)) +
  scale_color_d3() + scale_fill_d3()
```



Remodeling

After filtering our dataset and implementing changes to the naive model we arrive at a our new and improved model:

```
houses.glm2 <- glm(formula = Price~Bedrooms+Bathrooms+`Square Feet`+`Property Type`,
                   data=houses.df, family = Gamma(link="log"))
summary(houses.glm2)
```

```
##
## Call:
## glm(formula = Price ~ Bedrooms + Bathrooms + `Square Feet` +
##      `Property Type`, family = Gamma(link = "log"), data = houses.df)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1153  -0.3906  -0.1226   0.2059   3.7468
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.248e+01  2.009e-02  621.196 < 2e-16 ***
## Bedrooms       1.366e-02  6.007e-03   2.273  0.0230 *
## Bathrooms      8.720e-02  7.384e-03  11.809 < 2e-16 ***
## `Square Feet`  4.326e-04  9.736e-06  44.430 < 2e-16 ***
## `Property Type`Apt/Condo  2.023e-01  1.597e-02  12.669 < 2e-16 ***
## `Property Type`Townhouse  3.432e-02  1.628e-02   2.108  0.0351 *
## `Property Type`Duplex     1.920e-01  2.746e-02   6.993 2.84e-12 ***
## `Property Type`Mfd/Mobile Home -6.942e-01  6.089e-02 -11.402 < 2e-16 ***
## `Property Type`Multifamily  1.823e-02  6.005e-02   0.304  0.7615
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.2990719)
##
##      Null deviance: 5276.6  on 11215  degrees of freedom
## Residual deviance: 2736.3  on 11207  degrees of freedom
## AIC: 316241
##
## Number of Fisher Scoring iterations: 5
```